# Practice workbook · Kilobytes and kibibytes

Kilobyte vs kibibyte and the other prefixes; converting a stated quantity between decimal and binary multiples.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-16

# 1. Question 2 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

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**2** Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

**(2)**

| Denary | Hexadecimal |
|---|---|
| 8 | |
| 12 | |

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

**(1)**

- ☒ **A** $6^2$
- ☒ **B** $6 \times 2$
- ☒ **C** $6^2 - 1$
- ☒ **D** $2^6$

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

**(1)**

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

**(1)**

# 1. Question 2 · part e · 2 marks

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Worked-example retry · Attempt part e only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

**(2)**

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

**(2)**

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# 1. Question 2 · part e · 2 marks

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Worked-example retry · Attempt part e only.

(iii) Negating a signed integer means changing its sign without changing its value.

The negation of +16 is –16. The negation of –24 is +24.

Here is the binary bit pattern for a signed integer in two's complement format.

1110 0101

Convert the binary pattern to its negation in two's complement.

**(2)**

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(e) Construct an expression to convert 13 kilobytes to kibibytes.

You **do not** need to do the calculation.

**(2)**

**(Total for Question 2 = 13 marks)**

## 1. Question 2 · part e · 2 marks · Working and answer

Working and answer

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## 2. Question 4 · part a · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part a only.

**4** One function of an operating system is file management.

(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

**(2)**

(b) An image is 2322 pixels high and 4128 pixels wide.

The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

**(4)**

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## 2. Question 4 · part a · 2 marks · Working and answer

Working and answer

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# 3. Question 5 · part b.iv · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iv only.

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**5** Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

**(2)**

| Sensor | Purpose |
|---|---|
| Accelerometer | To measure body movement and step count |
| | |

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

**(2)**

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1 .....

.....

2 .....

.....

(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

**(1)**

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

# 3. Question 5 · part b.iv · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iv only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

**(2)**

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

**(2)**

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

**(1)**

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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P 7 8 6 3 7 A 0 1 4 2 0

## 3. Question 5 · part b.iv · 1 mark

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Further practice · Attempt part b.iv only.

(c) Wearable technology, such as smartwatches, has become very popular with many groups of people including athletes.

Discuss how the use of wearable technology can be a benefit to health.

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**(Total for Question 5 = 16 marks)**

## 3. Question 5 · part b.iv · 1 mark · Working and answer

Working and answer

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